

PHILIPPE WEIER

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📄 [Philippe Weier](#)

🎧 [WeiPhil](#)

Education

- 2021 – Now** **PhD Student**, Saarland University (Universität des Saarlandes), Germany
Computer Graphics Lab, advised by Prof. Dr.-Ing. Philipp Slusallek
Focus : Efficient Light Transport Simulation, Differentiable Rendering, Appearance Modelling
- 2017 – 2020** **Master in Computer Science Engineering**, EPFL, Switzerland.
Advanced Computer Graphics, Digital 3D Geometry Processing, Mathematical Foundation of Signal Processing, Distributed Algorithms, Signal Processing for Communications, Machine Learning, Cryptography and Security, Audio Signal Processing and Virtual Acoustics.
Thesis: Optimised Path Space Roughening
- 2016 – 2017** **3rd year Bachelor in Computer Science Engineering**, University of Granada, Spain
Computer Graphics, Operating Systems, Digital Image Processing, Software Engineering, Artificial Intelligence, Multiprocessor Architecture, Network Security.
- 2014 – 2016** **1st and 2nd year Bachelor in Computer Science Engineering**, EPFL, Switzerland.
Algorithmic, Computer Architecture, Functional Programming, Parallelism and Concurrency, Probability and Statistics, System Oriented Programming.

Experience

- 2025 June - Now** **Research Intern**, Meta, Zürich
Working on neural volumetric representations
Supervised by Sébastien Speierer and Adrian Jarabo
- 2024 April - Dec** **Research Intern**, Google, Zürich
Worked on physically-based inverse rendering
Supervised by Delio Vicini
- 2023 April - Sept** **Research Intern**, Adobe Systems, France
Worked on hybrid neural representation for physically-based rendering
Supervised by Élie Michel and Tamy Boubekeur
- 2022 May - Dec** **Research Intern**, Intel Graphics Lab, Germany
Worked on neural representation for appearance filtering and level of details
Supervised by Anton Kaplanyan and Lingqi Yan
- 2020 - 2021** **Rendering Researcher**, Weta Digital, New Zealand
Worked on the in-house physically-based renderer Manuka
- 2020 Feb - Aug** **Research Intern**, Weta Digital, New Zealand
Developed an improved formulation of roughening designed for production scale scenes in Manuka
Supervised by Johannes Hanika, Marc Droske and Wenzel Jakob
- 2019 Sep - Dec** **Research Student**, Realistic Graphics Laboratory (EPFL), Switzerland
Mesh-based pre-filtering of Complex Assets
Supervised by Guillaume Loubet and Wenzel Jakob
- 2019 Feb - Aug** **Research Intern**, Unity Labs Grenoble, France
Efficient Rendering of Anisotropic Layered Materials using an Atomic Decomposition with Statistical Operators
Supervised by Laurent Belcour
- 2017- 2019** **Teaching Assistant**, EPFL, Switzerland
Assisted the Professor with course, homeworks and exam material preparation in Computer Networks.

Publications

2025	Practical Inverse Rendering Of Textured And Translucent Appearance <i>ACM Transactions on Graphics (Proceedings of SIGGRAPH 2025)</i> Philippe Weier, Jérémy Riviere, Ruslan Guseinov, Stephan Garbin, Philipp Slusallek, Bernd Bickel, Thabo Beeler, Delio Vicini
2024	N-BVH: Neural ray queries with bounding volume hierarchies <i>ACM Transactions on Graphics (SIGGRAPH 2024 Conference Track)</i> Philippe Weier, Alexander Rath, Élie Michel, Iliyan Georgiev, Philipp Slusallek, Tamy Boubekeur
2023	Neural Prefiltering for Correlation-Aware Levels of Detail <i>ACM Transactions on Graphics (Proceedings of SIGGRAPH 2023)</i> Philippe Weier, Tobias Zirr, Anton Kaplanyan, Ling-Qi Yan, Philipp Slusallek
2022	EARS: Efficiency-Aware Russian Roulette and Splitting <i>ACM Transactions on Graphics (Proceedings of SIGGRAPH 2022)</i> Alexander Rath, Pascal Grittmann, Sebastian Herholz, Philippe Weier, Philipp Slusallek
2021	Optimised Path Space Regularisation <i>In Computer Graphics Forum (Proceedings of Eurographics Symposium on Rendering)</i> Philippe Weier, Marc Droske, Johannes Hanika, Andrea Weidlich, and Jiří Vorba.
2020	Rendering Layered Materials with Anisotropic Interfaces <i>In Journal of Computer Graphics Techniques</i> Philippe Weier, Laurent Belcour

Personal Projects

2023 - Now	NTWR: The Neural Twist Renderer A GPU physically-based renderer written in Cuda augmented with neural features
2020 - Now	Taranaki A physically-based toy renderer in Rust developped in my free time.
2019	Vulkan A GPU Oriented Prototyping tool in modern C++17 with a simple and flexible interface for more complex software or research validation tools.
2018	Procaryota A 2D space-shooter like game in C# made for fun in Unity.
2018	Lotr Army Companion An unofficial companion app for the table-top game Lord of The Rings. A full stack web application written in Javascript (React), MySQL and Python.

Skills

Programming

- C++ (Cuda, Vulkan)
- Rust
- Python
- C#, C (Embedded), Scala, Java
- Assembly (MIPS, ARM)
- Javascript (React), HTML, CSS, MySQL

Code Quality

- Continuous Integration, Tests
- Git, Docker

Languages

- French, Swiss German (Mother Tongues)
- Spanish (fluent)
- German (fluent)
- English (fluent)